# MOHAMED THARIK HUSSAIN

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**Github:** https://github.com/tharik-30

**Portfolio:** https://mohamedtharikhussain

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## PROFILE

Versatile B.E. Electronics & Communication Engineering graduate skilled in embedded hardware, firmware, and IoT system development. Proficient in microcontrollers, sensor integration, and PCB design, alongside web development skills in React, HTML, and CSS. Experienced in hardware testing, debugging, and cloud connectivity to ensure reliable, end-to-end smart solutions. Seeking an entry-level role to apply multi-disciplinary expertise across R&D, embedded engineering, and IoT development.

## EDUCATION

- **IFET COLLEGE OF ENGINEERING**

  *B.E in Electronics and Communication Engineering (2021-2025)*
  *Percentage: 7.06*
  *Villupuram*

- **ST.ANTONY'S HR SEC SCHOOL**

  *SSLC Year: 2019*
  *Percentage: 64*

  *HSC Year: 2021*
  *Percentage: 73.33*

## EXPERIENCE

- **IOT ENGINEER - HEPL**
  *Sept - 2026 to Present*

  As a Digital IT-IoT R&D Engineer, I design embedded hardware and firmware for connected milk analyzers. I program microcontrollers and sensors to measure fat, SNF, and density, while implementing IoT protocols for cloud data transmission, remote diagnostics, and safety compliance.

## CERTIFICATION

- Introduction Cyber security (Cisco)
- Cyber Thread management (Cisco)
- Networking Basics (Cisco)
- Endpoint Security (Cisco)
- Networking Devices (Cisco)
- Network Defence (Cisco)
- VLSI Course (Simplilearn)
- VLSI System on Chip Design (Maven Silicon)
- Electrical Engineer Job Simulation (GE Aerospace)
- Full Stack Development (Simplilearn)
- Embedded for Beginners - NIELIT Calicut
- EMBEDDED ERA - From Silicon to Smart Systems
- Data Visualisation: Empowering Business with effective Insights (TATA)

## PROJECT WORK

- **MOBILE PHONE DETECTOR**

  Cell phone detectors use two methods: detecting signal transmissions and sensing ferromagnetic components present in all mobile devices.

- **AUTONOMOUS DRONE USING IN DISASTERS (FINAL YEAR PROJECT)**

  Autonomous drones are rapidly deployed to earthquake sites, including zones with destroyed infrastructure, to independently transmit temporary telecommunication signals for emergency connectivity.
  **GitHub:**https://github.com/tharik-30/Drone-Final-Year-Project

- **MULTI-RAIL POWER SUPPLY UNIT**

  Designed a multi-output DC power supply system with 12V input, generating regulated rails of 7V, 5V, 3.3V, 2.5V, 1.8V, and 1.2V for embedded applications. Developed complete schematic, 4-layer PCB layout, and 3D board model using EasyEDA, following EMI-aware grounding, protection, and power-integrity practices.
  **GitHub:**https://github.com/tharik-30/multi-rail-power-supply-unit

## TECHNICAL SKILLS

- Internet of Things
- Embedded system
- C/C++
- Hardware Design
- MQTT
- PCB Design & Layout
- Soldering
- Microcontrollers
- Application Design

- **HARDWARE & ELECTRONICS:** Circuit Design, PCB Design & Layout, Electronic Component Selection, Power Supply Design, Hardware Debugging, Soldering.
- **EMBEDDED SYSTEMS:** Embedded C, C/C++, Microcontrollers, Arduino, STM32CubeIDE, IoT Fundamentals.
- **COMMUNICATION & INTERFACES:** UART, $I^2C$,$I^2S$,SPI
- **DESIGN & SIMULATION TOOLS:** Altium Designer, EasyEDA, VScode, MQTT Workbench.
- **PROGRAMMING & HDL:** C, C++, Embedded C, Python.
- **DEVELOPMENT TOOLS:** Git, GitHub, VS Code, Xilinx Vivado.

## LANGUAGE

- Tamil (Native)
- Hindi (Conversational)
- English (Proficiently)
- Arabic (Basic)

## ACHEIVEMENTS

- **BEST PROJECT AWARD – SOCIAL RELEVANCE (2025)**
  Awarded by IFET College of Engineering during the 27th Annual Day for presenting an innovative and socially impactful academic project.

## SOFT SKILLS

- Problem-Solving
- Technical Troubleshooting
- Teamwork and Collaboration
- Critical Thinking
- Time Management
- Debugging & Code Optimization

## HOBBIES

- Web Design & UI Exploration
- Photo and video Editing
- PC Gaming (Strategy & Simulation)
- Technology and Gadget Exploration
- Street Photography & Aesthetics
- Automobile Enthusiast